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During this project, I learned to design using a linked binary tree structure to represent Morse code. Each of the nodes has a letter, and a traversal based on Morse rules like “. “Is the left child and, “ _ “is the right child. The systems have a consist of three main components like TreeNode, the basic structre, MorseCodeTree, system that builds and manages the Morse tree, and MorseCodeConverter, this converts Morse into English. The algorithm for conversation is splitting input into words by using “/“, then splitng each word into letters by using spaces, then traversing the tree for each Morse code, then finaly append results into a string. My learning experience is learning to understand how trees can represent encoding systems, improving recursion skills, and undersetting the logic vs the UI. I assumed that the input contains a valid Morse code, that the only letters A-Z are being supported and that there is no punction required. The Junit test I have provideed each have a purpose. TestConvertSimpleString varifies that a basic morse code sentence can correctly tranalste into english. Then the testPrintTree checks that the morse code tree is built correctly.

The image displays two screenshots of a Java IDE's 'Run' console, showing the execution of JUnit tests for a Morse code conversion project.

The top screenshot shows the test run for `MorseCodeConverter_GFA_Test`. It indicates that 1 test passed (1 test total, 5 ms). The test is `testConvertToEnglishString`, which took 5 ms. The console output shows the Java command and the message "Process finished with exit code 0".

The bottom screenshot shows the test run for `MorseCodeConverterSimpleTest`. It indicates that 4 tests passed (4 tests total, 31 ms). The tests are `testConvertFile` (31 ms), `testPrintTree` (0 ms), `testConvertSimpleString` (0 ms), and `testConvertAnotherString` (0 ms). The console output shows the Java command and the message "Process finished with exit code 0".

